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Point2RBox-v2: Rethinking Point-supervised Oriented Object Detection with Spatial Layout Among Instances

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https://github.com/VisionXLab/point2rbox-v2

Abstract

With the rapidly increasing demand for oriented object detection (OOD), recent research involving weakly-supervised detectors for learning OOD from point annotations has gained great attention. In this paper, we rethink this challenging task setting with the layout among instances and present Point2RBox-v2. At the core are three principles: ***1) Gaussian overlap loss.*** *It learns an upper bound for each instance by treating objects as 2D Gaussian distributions and minimizing their overlap.* ***2) Voronoi watershed loss.*** *It learns a lower bound for each instance through watershed on Voronoi tessellation.* ***3) Consistency loss.*** *It learns the size/rotation variation between two output sets with respect to an input image and its augmented view. Supplemented by a few devised techniques, e.g. edge loss and copy-paste, the detector is further enhanced. To our best knowledge, Point2RBox-v2 is the first approach to explore the spatial layout among instances for learning point-supervised OOD. Our solution is elegant and lightweight, yet it is expected to give a competitive performance especially in densely packed scenes: 62.61%/86.15%/34.71% on DOTA/HRSC/FAIR1M.*

1. Introduction

Emerged as an essential task in computer vision, oriented object detection (OOD) has become a prominent demand in autonomous driving [6], aerial images [7, 30, 48, 51, 52], scene text [23, 28, 72], retail scenes [10, 35], industrial inspection [29, 47], and more.

Manual annotations are essential to teach the detector new concepts of visual objects. Early research is often supervised by rotated bounding boxes (RBoxes), where annotations align with the desired output. While this approach has demonstrated promising results, RBox labels are considerably expensive. The cost of annotating each RBox is approximately 36.5% higher than a horizontal bounding box (HBox) and 104.8% higher than a point [66]. This highlights the importance of utilizing coarser labels for weakly-supervised oriented detection. Recent advancements in HBox-supervised OOD, particularly with H2RBox [59] and H2RBox-v2 [65], have shown promise in bridging the gap between HBox- and RBox-supervised methods, thereby lessening the dependence on labor-intensive RBox labeling.

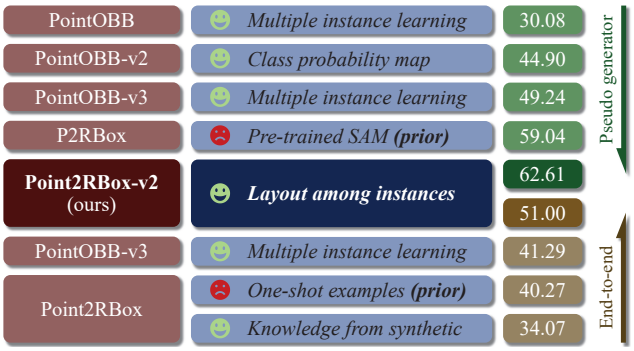


Figure 1. Related methods, their principles for knowledge mining, whether using additional priors, and performance on DOTA-v1.0.

Following HBox-supervised methods, the more challenging point-supervised OOD opens up a new avenue. Several innovative approaches have emerged in the past year (see Fig. 1): **1) Point-prompt OOD.** P2RBox [2], PMHO [69], and PointSAM [27] employ the zero-shot ability of SAM [17]. **2) Pseudo generation.** PointOBB [33] and its v2/v3 version [38, 68] use multiple instance learning and class probability map for RBox generation. **3) Knowledge combination.** Point2RBox [66] learns from one-shot examples. However, existing methods overlook the spatial relationships between objects. This brings us to the motivation of this paper.

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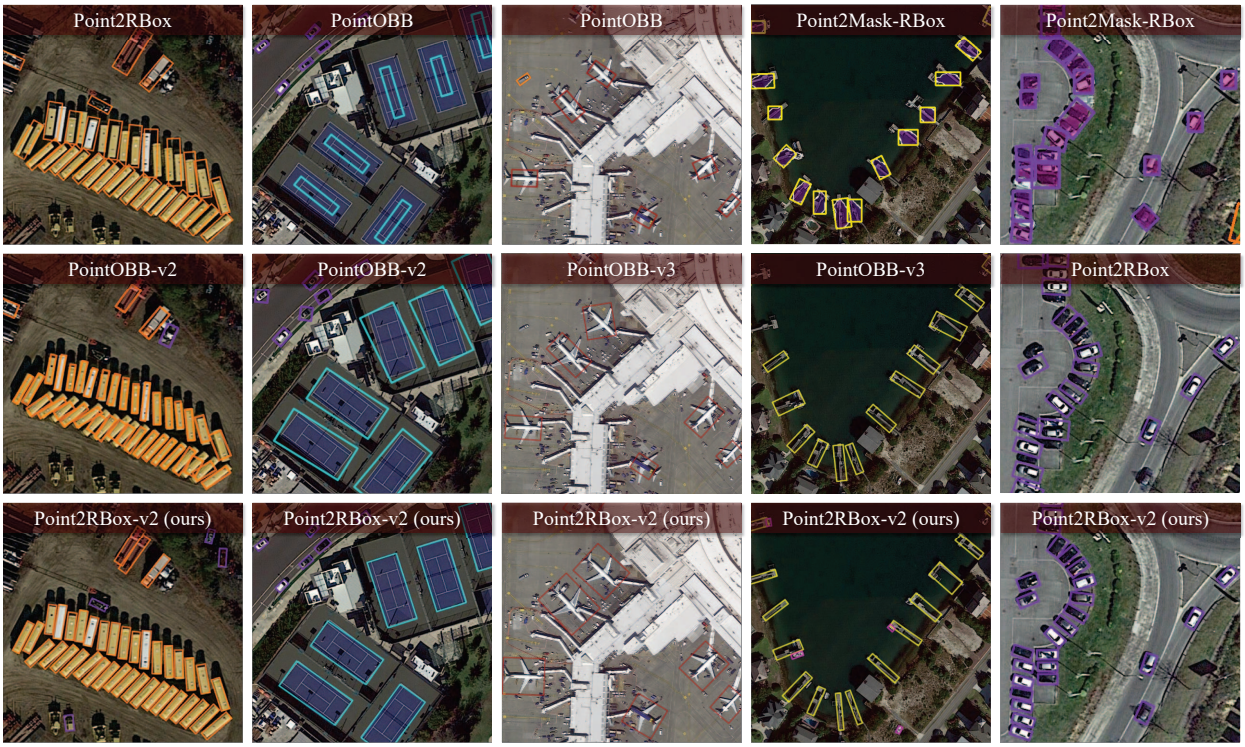


Figure 2. Visual comparisons with state-of-the-art methods including: Point2Mask (2023) [21], PointOBB (2024) [33], PointOBB-v2 (2025) [38], PointOBB-v3 (2025) [68], and Point2RBox (2024) [66]. The boxes detected by our method (last row) wrap the objects more tightly.

Motivation. While point-supervised OOD has gained attention, the utilization of relationships among instances remains absent in the literature. For example, the arrangement of vehicles in a parking lot can serve as effective constraints for learning their size and orientation. This approach may yield valuable insights into point-supervised OOD, particularly in densely packed scenes where current methods face significant challenges. Can we leverage the spatial layout of objects to enhance point-supervised OOD? In this paper, we dive into this idea and seek to answer this question.

Highlights. **1)** Point2RBox-v2 is proposed for point-supervised OOD, advancing the state of the art as displayed in Fig. 2 and Tables 1-2. **2)** We propose novel and elegant losses to enforce constraints from the spatial layout among instances based on Gaussian overlap and Voronoi tessellation [1]. **3)** Other modules are devised (i.e. edge loss, consistency loss, copy-paste) to further enhance the method.

Contributions. **1)** To our best knowledge, this work is the first attempt to learn point-supervised OOD from the layout among objects, where we propose novel principles based on Gaussian overlap and Voronoi tessellation. **2)** The training pipeline and detailed implementation are elucidated, with necessary modules (i.e. edge loss, consistency loss, copy-paste) incorporated. The source code will be made publicly available. **3)** Extensive experiments demonstrate that leveraging the spatial layout of instances can significantly advance the state of the art, surpassing other alternatives in accuracy.

2. Related Work

2.1. RBox-supervised Oriented Detection

In addition to horizontal detection [26, 71], oriented object detection (OOD) [46, 52] has received extensive attention. Representative works include anchor-based detector Rotated RetinaNet [25], anchor-free detector Rotated FCOS [42], and two-stage solutions, e.g. RoI Transformer [5], Oriented R-CNN [49], and ReDet [11]. Some research enhances the detector by exploiting alignment features, e.g. R^3Det [55] and S^2A-Net [12]. The angle regression may face boundary discontinuity and remedies are developed, including modulated losses [37, 53, 58] that alleviate loss jumps, angle coders [50, 51, 54, 63] that convert the angle into boundary-free coded data, and Gaussian-based losses [34, 56, 57, 60, 61] transforming rotated bounding boxes into Gaussian distributions. RepPoint-based methods [15, 20, 62] provide alternatives that predict a set of points that bounds the spatial extent of an object. LMMRotate [19] is a new paradigm of OOD based on multimodal language model.

2.2. Point-supervised Oriented Detection

Recently, several methods for point-supervised OOD have been proposed: **1)** P2RBox [2], PMHO [69], and PointSAM [27] propose point-prompt OOD by employing the zero-shot Point-to-Mask ability of SAM [17]. **2)** Point2RBox [66] introduces an end-to-end approach based on knowledge com-

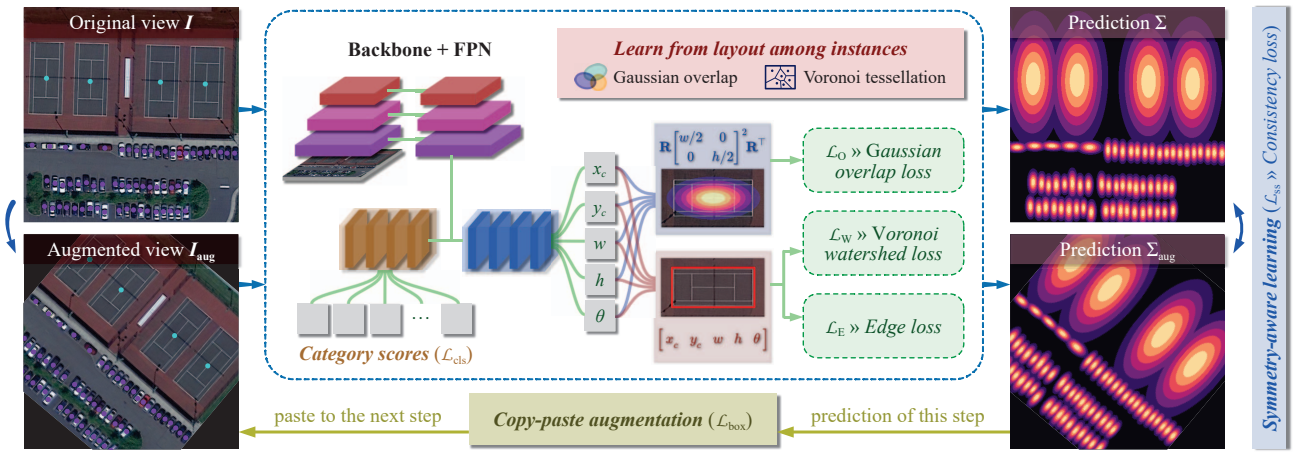


Figure 3. The training pipeline of Point2RBox-v2. Gaussian overlap loss and Voronoi watershed loss utilize the spatial layout (see Fig. 4), while edge loss (see Sec. 3.4), symmetry-aware learning (see Sec. 3.5), and copy-paste (see Sec. 3.6) further enhance the method.

bination. **3)** PointOBB [33, 68] achieves RBox generation through multiple instance learning. **4)** PointOBB-v2 [38] learns a class probability map to generate pseudo RBoxes. **5)** Wholly-WOOD [67] uses points, HBoxes, RBoxes, or their combinations for mixed supervised OOD.

Among these methods, P2RBox, PMHO, and PointSAM rely on the SAM model pre-trained on massive labeled datasets, whereas Point2RBox requires one-shot examples for each category. In contrast, PointOBB series do not use many priors, but they necessitate two-stage training.

2.3. Other Weakly-supervised Settings

Compared to Point-to-RBox, some other settings have been better studied. These methods are potentially applicable to our Point-to-RBox task setting by using a cascade pipeline, such as Point-to-HBox-to-RBox. In our experiment, cascade pipelines powered by state-of-the-art approaches are also compared. Here, representative works are introduced.

HBox-to-RBox. H2RBox [59] establishes a paradigm that limits the object to a few candidate angles through geometric constraint from HBoxes, with a self-supervised branch eliminating the undesired results. An enhanced version H2RBox-v2 [65] is proposed to leverage the reflection symmetry of objects to further boost the accuracy. EIE-Det [45] uses an explicit equivariance branch for learning rotation consistency, and an implicit equivariance branch for learning position, aspect ratio, and scale consistency. KCR [74] combines RBox- and HBox-annotated datasets for transfer learning. Some studies [16, 41] use additional annotated data for training, which are also attractive but less general.

Point-to-HBox. Several related approaches have been developed, including: **1)** P2BNet [3] samples box proposals of different sizes around the labeled point and classify them to achieve point-supervised horizontal object detection. **2)** PSOD [8] achieves point-supervised salient object detection using an edge detector and adaptive masked flood fill.

Point-to-Mask. Point2Mask [21] is proposed to achieve panoptic segmentation using single point annotation per target. SAM (Segment Anything Model) [17] produces object masks from input point/HBox prompts. Though RBoxes can be obtained from the segmentation mask by finding the minimum circumscribed rectangle, such a complex pipeline can be less cost-efficient and perform worse [59, 65].

3. Method

3.1. Overview and Preliminary

An overview of Point2RBox-v2 is illustrated in Fig. 3. The network is based on ResNet50 [13] backbone, FPN [24] head, and PSC [64] angle coder. Objects of varying sizes are typically assigned to different FPN layers based on the scale. However, points lack size, so we assign them all to the P3 layer (stride = 8). Assume the detector $f_{\text{nn}}(\cdot)$ maps an image I to a set of RBoxes as detection results:

$$(x_c, y_c, w, h, \theta) = f_{\text{nn}}(I) \tag{1}$$

Equivalently, oriented objects can also be represented by 2D Gaussian distributions $\mathcal{N}(\mu, \Sigma)$ [60]:

$$\left\{ \mu = \begin{bmatrix} x_c & y_c \end{bmatrix}, \quad \Sigma = \mathbf{R} \begin{bmatrix} w/2 & 0 \\ 0 & h/2 \end{bmatrix}^2 \mathbf{R}^\top \right\} \tag{2}$$

where

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{3}$$

At the core of Point2RBox-v2 is to utilize the constraints from the layout, which is achieved by Gaussian overlap loss (Sec. 3.2) and Voronoi watershed loss (Sec. 3.3). These two losses effectively limit the size and rotation of objects to a reasonable range. Upon that, edge loss (Sec. 3.4) aligns the bounding box with the edge of objects to improve the accuracy. Incorporated with symmetry-aware learning (Sec. 3.5) and copy-paste augmentation (Sec. 3.6), we achieve a

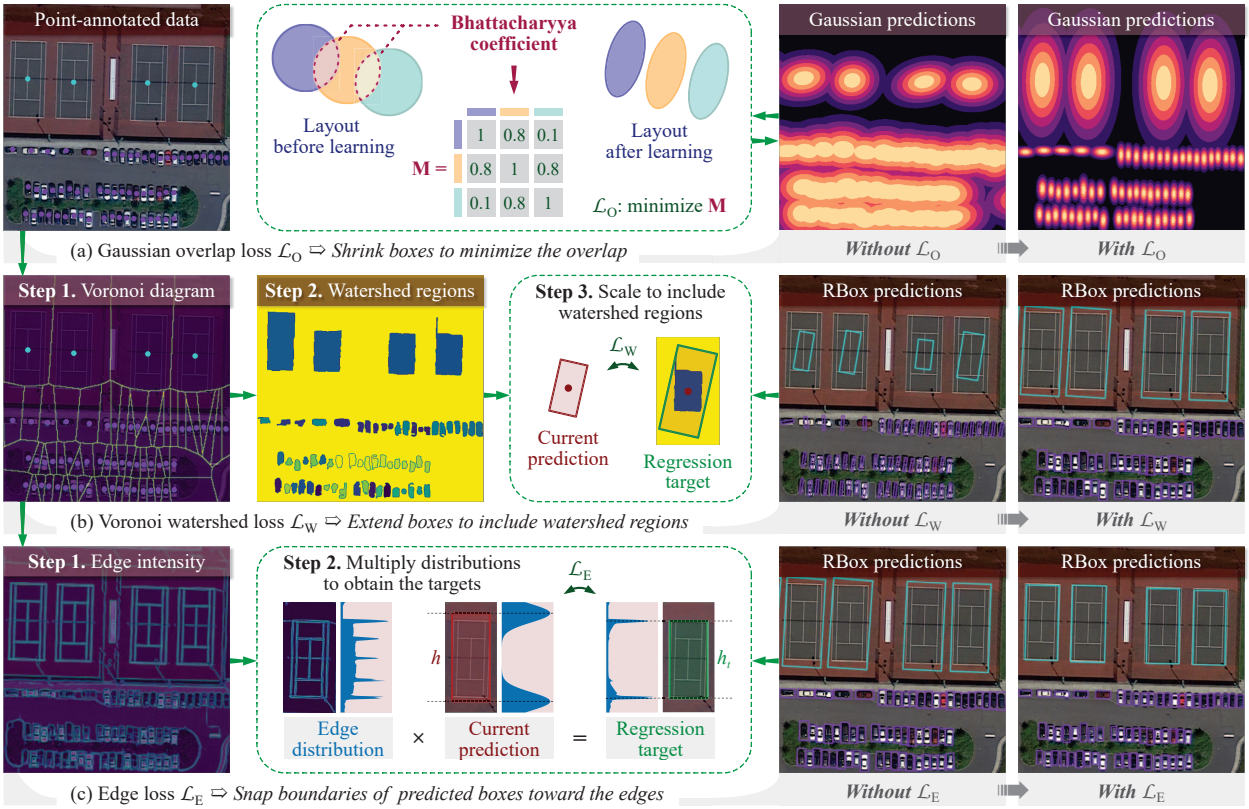


Figure 4. To illustrate the procedure of the three newly proposed loss functions and their impact on the learning results. (a) Gaussian overlap loss (see Sec. 3.2). (b) Voronoi watershed loss (see Sec. 3.3). (c) Edge loss (see Sec. 3.4).

stable and high-accuracy solution for point-supervised OOD. In subsequent subsections, these modules are detailed.

3.2. Gaussian Overlap Loss

As mentioned, oriented objects can be represented by 2D Gaussian distributions $\mathcal{N}(\mu, \Sigma)$. The overlap volume between two distributions $\mathcal{N}_1(\mu_1, \Sigma_1)$ and $\mathcal{N}_2(\mu_2, \Sigma_2)$ can be approximated by the Bhattacharyya coefficient [60] as:

$$B\left(\mathcal{N}_1, \mathcal{N}_2\right) = \exp\left(-\frac{1}{8}\mu^\top \Sigma^{-1}\mu\right) \cdot \frac{|\Sigma_1|^{1/4}|\Sigma_2|^{1/4}}{|\Sigma|^{1/2}} \quad (4)$$

where $\mu = \mu_2 - \mu_1$, $\Sigma = \frac{1}{2}(\Sigma_1 + \Sigma_2)$, $|\Sigma|$ denotes the determinant of the covariance matrix.

Based on the above equation, we build a Gaussian overlap matrix $\mathbf{M} \in \mathbb{R}^{N\times N}$ for each scene image as:

$$\mathbf{M}_{i,j} = B\left(\mathcal{N}_i, \mathcal{N}_j\right) \quad (5)$$

where $i, j = 1, 2, \ldots, N$; N is the distribution count (or the instance count) within one training image.

The Gaussian overlap loss can then be expressed as:

$$\mathcal{L}_O = \frac{1}{N}\sum_{i\neq j}\left(\mathbf{M}_{i,j}\right) \quad (6)$$

where $i \neq j$ omits diagonal elements. With this loss, the detector learns to arrange instances (see Fig. 4a) based on the mutual exclusivity among instances.

3.3. Voronoi Watershed Loss

A Voronoi diagram [1] is a partitioning of a space based on a set of points. Point annotations can be effectively utilized to calculate a Voronoi diagram, where a distinct polygon region is assigned to each point-annotated instance. The watershed algorithm [44], on the other hand, is a region segmentation technique that treats the intensity of pixels as a topographic surface, identifying regions as "catchment basins".

The calculation of Voronoi ridges can be formulated as:

$$V = Voronoi\left(X\right) \quad (7)$$

where X are the annotated points within a training image; V are the output Voronoi ridges (pixel coordinates).

Interestingly, we find that Voronoi diagrams can be utilized as initial markers for watershed to obtain a region for each instance (see Fig. 4b). In concrete terms, the points X can be employed as foreground markers, while the Voronoi ridges V can act as background boundaries:

$$S = Watershed\left(I, X, V\right) \quad (8)$$

where S are the output basin regions (pixel coordinates) corresponding to each annotated instance. By rotating S

to align with the direction of the current prediction, the regression target of width and height can be expressed as:

$$\begin{bmatrix} w_t \\ h_t \end{bmatrix} = 2 \max \left| \mathbf{R}^\top \left(S - \begin{bmatrix} x_c \\ y_c \end{bmatrix} \right) \right| \tag{9}$$

where (x_c, y_c, w, h, θ) is the current prediction; $\mathbf{R}$ is defined by Eq. (3); w_t and h_t are detached to stop the gradient.

Afterward, the Voronoi watershed loss to regress the width and height of objects can be calculated as:

$$\mathcal{L}_W = L_{\text{GWD}} \left(\begin{bmatrix} w/2 & 0 \\ 0 & h/2 \end{bmatrix}^2, \begin{bmatrix} w_t/2 & 0 \\ 0 & h_t/2 \end{bmatrix}^2 \right) \tag{10}$$

where $L_{\text{GWD}}(\cdot)$ is Gaussian Wasserstein Distance Loss [60].

3.4. Edge Loss

The above two losses have limited the size to a reasonable range. To make it more accurate, we propose the edge loss to snap the boundaries toward the edges (see Fig. 4c).

First, a region $P \in \mathbb{R}^{(2K+1)\times(2K+1)}$ around each predicted RBox is extracted via Rotated RoI Align [14]:

$$P = RoIAlign\left(E\left(I\right), (x_c, y_c, \beta w, \beta h, \theta)\right) \tag{11}$$

where $E(\cdot)$ is the edge detection function [39]. We set $K = 24$ and $\beta = 1.6$ in our experiments (see Table 9).

By calculating the sum of each row of P, the edge distribution in y direction can be obtained as:

$$\mu_i = \sum_{j=1}^{2K+1} \left(P_{(K+1-i),j} + P_{(K+1+i),j} \right) \tag{12}$$

where $i = 1, 2, \ldots, K$, indicating that the upper half of P is reversed and added to the lower half.

Meanwhile, the current prediction of the edge can also be softened into a distribution as:

$$\lambda_i = \exp\left(-\frac{(i - K/\beta)^2}{2{\sigma_E}^2} \right) \tag{13}$$

where $i = 1, 2, \ldots, K$; σ_E is set to 6. Note that we crop P based on the predicted RBox, thus λ is always the same.

Multiplying the two distributions yields the target:

$$h_t = \frac{\beta h}{K} \arg\max(\mu \times \lambda) \tag{14}$$

where h_t is the regression target of height. Likewise, the width target w_t is calculated along x direction.

The edge loss is then obtained as:

$$\mathcal{L}_E = smooth_{L1}\left(\begin{bmatrix} w & h \end{bmatrix}, \begin{bmatrix} w_t & h_t \end{bmatrix} \right) \tag{15}$$

Note that $\mathcal{L}_W$ and $\mathcal{L}_E$ merely refines the width and height of boxes, without involving the angle regression.

3.5. Symmetry-aware Learning

Proven effective in learning the rotation of objects [65], in this work, symmetry-aware learning is extended to Gaussian-based OOD. In the left part of Fig. 3, the training image I is transformed (randomly selected from rotation, flip, and scale, see Table 7) to generate an augmented view as:

$$I_{\text{aug}} = \alpha I \tag{16}$$

where

$$\alpha = \begin{bmatrix} \cos\mathcal{R} & -\sin\mathcal{R} \\ \sin\mathcal{R} & \cos\mathcal{R} \end{bmatrix}^{p_1} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}^{p_2} \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}^{p_3} \tag{17}$$

where $(p_1, p_2, p_3) = (1, 0, 0)$ when rotation is selected, $(0, 1, 0)$ when flip, $(0, 0, 1)$ when scale. $\mathcal{R}$ and s are the random amount of rotation and scale, $s \in (0.5, 0.9)$.

By feeding both I and I_{aug} into the network, we obtain two sets of output Gaussian distributions and angles:

$$\begin{cases} (\Sigma, \theta) = f_{\text{nn}}(I) \\ (\Sigma_{\text{aug}}, \theta_{\text{aug}}) = f_{\text{nn}}(I_{\text{aug}}) \end{cases} \tag{18}$$

The consistency loss is calculated between the two sets so that the network learns the size and rotation variation:

$$\mathcal{L}_{\text{ss}} = L_{\text{GWD}}\left(\alpha\Sigma\alpha^\top, \Sigma_{\text{aug}}\right) + L_{\text{ANG}}\left(m\theta + \mathcal{R}, \theta_{\text{aug}}\right) \tag{19}$$

where $\mathcal{R}$ is the rotation angle and $m = 1$ when rotation is selected; $(m, \mathcal{R}) = (-1, 0)$ when flip; $(m, \mathcal{R}) = (1, 0)$ when scale. $L_{\text{GWD}}(\cdot)$ is Gaussian Wasserstein Distance Loss [60] and $L_{\text{ANG}}(\cdot)$ is defined as:

$$L_{\text{ANG}}(\theta_1, \theta_2) = \min_{k \in Z} \left(smooth_{L1}\left(\theta_1, k\pi + \theta_2\right) \right) \tag{20}$$

where $\min(\cdot)$ regresses the prediction toward the closest target to circumvent the periodicity problem [65].

3.6. Copy-paste Augmentation

Inspired by [9], we propose to crop the detected instances of step k and paste them on the training image of step $k+1$. The maximum number of paste boxes is limited to 10 in each step. We simply use the bounding boxes of the cropped instances as the regression targets, and use Gaussian Wasserstein Distance Loss [60] to calculate $\mathcal{L}_{\text{box}}$.

3.7. Overall Loss

The overall loss $\mathcal{L}$ for Point2RBox-v2 can be expressed as:

$$\mathcal{L}_{\text{cls}} + w_{\text{box}}\mathcal{L}_{\text{box}} + w_{\text{O}}\mathcal{L}_{\text{O}} + w_{\text{W}}\mathcal{L}_{\text{W}} + w_{\text{E}}\mathcal{L}_{\text{E}} + w_{\text{ss}}\mathcal{L}_{\text{ss}} \tag{21}$$

where $\mathcal{L}_{\text{cls}}$ is the focal loss [25] for classification, $\mathcal{L}_{\text{box}}$ regresses boxes/centers toward copy-paste/ground-truth labels, w_{box} is set to one by default, $(w_{\text{O}}, w_{\text{W}}, w_{\text{E}}, w_{\text{ss}})$ are set to $(10, 5, 0.3, 1)$ based on our ablation studies (see Tables 3-5).

4. Experiments

Experiments are carried out on NVIDIA RTX4090 GPUs using PyTorch 2.2.0 [36] and the rotation detection tool kits: MMRotate 1.0.0 [73]. All the experiments follow the same hyper-parameters (learning rate, batch size, optimizer, etc.).

Methods	*	PL[1]	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}
▼ *RBox-supervised OOD*																	
RepPoints (2019) [62]	✓	86.7	81.1	41.6	62.0	76.2	56.3	75.7	90.7	80.8	**85.3**	**63.3**	66.6	59.1	67.6	33.7	68.45
RetinaNet (2017) [25]	✓	88.2	77.0	45.0	69.4	71.5	59.0	74.5	90.8	**84.9**	79.3	57.3	64.7	62.7	66.5	39.6	68.69
GWD (2021) [56]	✓	**89.3**	75.4	47.8	61.9	79.5	73.8	86.1	**90.9**	84.5	79.4	55.9	59.7	63.2	71.0	45.4	71.66
FCOS (2019) [42]	✓	89.1	76.9	50.1	63.2	**79.8**	**79.8**	87.1	90.4	80.8	84.6	59.7	66.3	65.8	71.3	41.7	72.44
S^2A-Net (2022) [12]	✓	89.2	**83.0**	**52.5**	**74.6**	78.8	79.2	**87.5**	**90.9**	**84.9**	84.8	61.9	**68.0**	**70.7**	**71.4**	**59.8**	**75.81**
▼ *HBox-supervised OOD*																	
Sun et al. (2021) [41]	×	51.5	38.7	16.1	36.8	29.8	19.2	23.4	83.9	50.6	80.0	18.9	50.2	25.6	28.7	25.5	38.60
BoxInst-RBox (2021) [43][2]	×	68.4	40.8	33.1	32.3	46.9	55.4	56.6	79.5	66.8	82.1	41.2	52.8	52.8	65.0	30.0	53.59
H2RBox (2023) [59]	✓	88.5	73.5	40.8	56.9	77.5	65.4	77.9	**90.9**	83.2	**85.3**	55.3	62.9	52.4	63.6	43.3	67.82
EIE-Det (2024) [45]	✓	87.7	70.2	41.5	**60.5**	**80.7**	**76.3**	86.3	**90.9**	82.6	84.7	53.1	**64.5**	58.1	70.4	43.8	70.10
H2RBox-v2 (2023) [65]	✓	**89.0**	**74.4**	**50.0**	**60.5**	79.8	75.3	**86.9**	**90.9**	**85.1**	85.0	**59.2**	63.2	**65.2**	**70.5**	**49.7**	**72.31**
▼ *Point-supervised OOD*																	
Point2Mask-RBox (2023) [21][2]	×	4.0	23.1	3.8	1.3	15.1	1.0	3.3	19.0	1.0	29.1	0.0	9.5	7.4	21.1	7.1	9.72
P2BNet+H2RBox (2023) [3, 59]	×	24.7	35.9	7.1	27.9	3.3	12.1	17.5	17.5	0.8	34.0	6.3	49.6	11.6	27.2	18.8	19.63
P2BNet+H2RBox-v2 (2023) [3, 65]	×	11.0	44.8	14.9	15.4	36.8	16.7	27.8	12.1	1.8	31.2	3.4	50.6	12.6	36.7	12.5	21.87
P2RBox (2024) [2]†	×	87.8	65.7	15.0	**60.7**	73.0	71.7	78.9	81.5	44.5	81.2	**41.2**	39.3	45.5	57.5	41.2	59.04
PointOBB (2024) [33]	×	26.1	65.7	9.1	59.4	65.8	34.9	29.8	0.5	2.3	16.7	0.6	49.0	21.8	41.0	36.7	30.08
Point2RBox (2024) [66]	✓	62.9	64.3	14.4	35.0	28.2	38.9	33.3	25.2	2.2	44.5	3.4	48.1	25.9	45.0	22.6	34.07
Point2RBox+SK (2024) [66]†	✓	53.3	63.9	3.7	50.9	40.0	39.2	45.7	76.7	10.5	56.1	5.4	49.5	24.2	51.2	33.8	40.27
Point2RBox+SK (2024) [66]†	×	66.4	59.5	5.2	52.6	54.1	53.9	57.3	**90.8**	3.2	57.8	6.1	47.4	22.9	55.7	40.5	44.90
PointOBB-v2 (2025) [38]	×	64.5	27.8	1.9	36.2	58.8	47.2	53.4	90.5	62.2	45.3	12.1	41.7	8.1	43.7	32.0	41.68
PointOBB-v3 (2025) [68]	✓	30.9	39.4	13.5	22.7	61.2	7.0	43.1	62.4	59.8	47.3	2.7	45.1	16.8	55.2	11.4	41.29
PointOBB-v3 (2025) [68]	×	52.9	54.4	**21.3**	52.7	65.6	44.9	67.8	87.2	26.7	73.4	32.6	**53.3**	39.0	56.4	10.2	49.24
Point2RBox-v2 (ours)	✓	78.4	52.7	8.3	40.9	71.0	60.5	74.7	88.7	65.5	72.1	24.4	26.1	30.1	50.7	21.0	51.00
Point2RBox-v2 (ours)	×	**88.0**	**72.6**	8.0	46.2	**79.6**	**76.3**	**86.9**	89.1	**79.7**	**82.9**	26.2	45.3	**45.8**	**66.3**	**46.3**	**62.61**

*Comparison tracks: ✓= End-to-end training and testing; × = Generating pseudo labels to train the FCOS detector (two-stage training).
†Using additional priors. P2RBox: Pre-trained SAM model; Point2RBox+SK: One-shot sketches for each class.
[1]PL: Plane, BD: Baseball diamond, BR: Bridge, GTF: Ground track field, SV: Small vehicle, LV: Large vehicle, SH: Ship, TC: Tennis court, BC: Basketball court, ST: Storage tank, SBF: Soccer-ball field, RA: Roundabout, HA: Harbor, SP: Swimming pool, HC: Helicopter.
[2]-RBox: The minimum rectangle operation is performed on the output Mask to obtain the RBox.

Table 1. Detection performance of each category on the DOTA-v1.0 and the mean AP_{50} of all categories.

Average precision (AP) is adopted as the primary metric. All the models are configured upon ResNet50 [13] and trained with AdamW [31]. **1) Learning rate.** Initialized at 5e-5, warm-up for 500 iterations, and divided by ten at each decay step. **2) Epochs.** 72 for HRSC; 12 for the others. **3) Augmentation.** Random rotation/flip for HRSC; random flip for the others. **4) Image size.** Split into 1,024 × 1,024 with an overlap of 200 for DOTA/FAIR1M/STAR; scaled to 800 × 800 for others. **5) Multi-scale.** All experiments evaluated without multi-scale technique [73]. **6) Datasets.** Six remote sensing and one retail scene datasets, covering all datasets used by the main counterparts [2, 33, 66]:

- **DOTA [48].** DOTA-v1.0 has 2,806 aerial images annotated with 15 categories, while DOTA-v1.5/2.0 are the extended versions with more small objects and categories.
- **DIOR [4].** It is an aerial image dataset re-annotated with RBoxes based on its original HBox version [18], with a high variation in object size and high intra-class diversity.
- **HRSC [30].** It contains ship instances on the sea and inshore. The train/val/test set includes 436/181/444 images.
- **FAIR1M [40].** It has more than 1 million instances and more than 40,000 images for fine-grained object recognition in remote sensing imagery, annotated with 37 categories. The results are evaluated on FAIR1M-1.0.
- **STAR [22].** It is extensive for scene graph generation, covering more than 210,000 objects with diverse spatial resolutions, classified into 48 fine-grained categories and precisely annotated with oriented bounding boxes.
- **SKU110K [35].** It focuses on the detection of densely packed retail scenes with 110,712 objects in 11,762 images. The density reaches 86 instances per image.
- **RSAR [70].** It is a remote sensing dataset based on Synthetic Aperture Radar (SAR) imagery with 6 categories.

4.1. Main Results on DOTA-v1.0

Table 1 compares Point2RBox-v2 with the state-of-the-art methods, which can be categorized into two tracks:

1) End-to-end training. These methods apply the trained weakly-supervised detector directly to the test set. Without relying on priors, our approach demonstrates an improvement of 16.93% (51.00% vs. 34.07%) compared to Point2RBox. Even when compared to Point2RBox+SK, which incorporates additional data-side priors (i.e. one-shot examples for each class), our method still outperforms it by 10.73% (51.00% vs. 40.27%).

2) Two-stage training. These methods generate RBox

Methods	*	DOTA-v1.0	DOTA-v1.5	DOTA-v2.0	DIOR	HRSC	FAIR1M	STAR	SKU110K	RSAR
▼ *RBox-supervised OOD*										
RetinaNet (2017) [25]	✓	68.69	60.57	47.00	54.96	84.49	37.67	21.80	78.50	57.67
GWD (2021) [56]	✓	71.66	63.27	48.87	57.60	86.67	39.11	25.30	79.16	57.80
FCOS (2019) [42]	✓	72.44	64.53	51.77	59.83	88.99	41.25	**28.10**	80.09	**66.66**
S^2A-Net (2022) [12]	✓	**75.81**	**66.53**	**52.39**	**61.41**	**90.10**	**42.44**	27.30	**80.36**	66.47
▼ *HBox-supervised OOD*										
Sun et al. (2021) [41]	×	38.60	-	-	-	-	-	-	-	-
KCR (2023) [74]	✓	-	-	-	-	79.10	-	-	-	-
H2RBox (2023) [59]	✓	70.05	61.70	48.68	57.80	7.03	35.94	17.20	57.15	49.92
H2RBox-v2 (2023) [65]	✓	72.31	64.76	50.33	57.64	**89.66**	**42.27**	**27.30**	**70.70**	**65.16**
AFWS (2024) [32]	✓	**72.55**	**65.92**	**51.73**	**59.07**	-	41.80	-	-	-
▼ *Point-supervised OOD*										
P2RBox (2024) [2]†	×	59.04	-	-	-	-	-	-	-	-
PointSAM (2025) [27]†	×	-	-	-	**46.20**	-	-	-	-	-
PointOBB (2024) [33]	×	30.08	10.66	5.53	37.31	-	11.19	9.19	-	13.80
Point2RBox+SK (2024) [66]†	✓	40.27	30.51	23.43	27.34	79.40	20.03	7.86	3.41	27.81
PointOBB-v2 (2025) [38]	×	41.68	30.59	20.64	39.56	-	13.36	9.00	56.63	18.99
PointOBB-v3 (2025) [68]	✓	41.20	31.25	22.82	37.60	-	11.42	11.31	-	15.84
PointOBB-v3 (2025) [68]	×	49.24	33.79	23.52	40.18	-	18.35	12.85	-	22.60
Point2RBox-v2 (ours)	✓	51.00	39.45	27.11	34.70	82.67	25.72	7.80	64.00	28.60
Point2RBox-v2 (ours)	×	**62.61**	**54.06**	**38.79**	44.45	**86.15**	**34.71**	**14.20**	**65.64**	**30.90**

*Comparison tracks: ✓= End-to-end training and testing; × = Generating pseudo labels to train the FCOS detector (two-stage training).
†Using additional priors. P2RBox/PointSAM: Pre-trained SAM model; Point2RBox+SK: One-shot sketches for each class.

Table 2. Accuracy (AP_{50}) comparisons on the DOTA-v1.0/1.5/2.0, DIOR, HRSC, FAIR1M, STAR, SKU110K, and RSAR datasets.

w_O	DOTA		HRSC	
	E2E	FCOS	E2E	FCOS
3	48.76	61.62	81.85	84.36
5	49.81	62.44	82.46	85.76
10	**51.00**	**62.61**	**82.67**	**86.15**
30	45.88	57.83	81.56	85.61

Table 3. Ablation with the weight of $\mathcal{L}_O$.

w_W	DOTA		HRSC	
	E2E	FCOS	E2E	FCOS
3	50.85	56.78	78.42	83.49
5	**51.00**	**62.61**	**82.67**	**86.15**
10	49.15	60.54	30.37	35.13
30	42.84	52.53	23.89	25.91

Table 4. Ablation with the weight of $\mathcal{L}_W$.

w_E	DOTA		HRSC	
	E2E	FCOS	E2E	FCOS
0.1	48.75	57.62	34.71	39.45
0.3	51.00	62.61	**82.67**	**86.15**
0.5	**51.36**	**62.63**	76.85	85.22
1.0	49.05	60.63	56.59	59.59

Table 5. Ablation with the weight of $\mathcal{L}_E$.

w_{ss}	DOTA		HRSC	
	E2E	FCOS	E2E	FCOS
0.1	49.28	59.66	73.66	78.92
1.0	**51.00**	**62.61**	**82.67**	**86.15**
3.0	49.15	59.20	1.30	1.65

Table 6. Ablation with the weight of $\mathcal{L}_{ss}$.

R / F / S	DOTA	HRSC	R / F / S	DOTA	HRSC
90% / 10% / 0%	60.42	85.46	80% / 20% / 0%	59.46	84.73
75% / 0% / 25%	60.79	86.22	60% / 15% / 25%	62.38	84.21
68% / 7% / 25%	**62.61**	**86.15**	38% / 37% / 25%	45.87	8.56
45% / 5% / 50%	60.55	85.34	40% / 10% / 50%	60.49	10.74

Table 7. Ablation with the proportion of augmented views in self-supervision.

labels on train/val sets, with which the FCOS detector is trained. In this two-stage mode, Point2RBox-v2 achieves an accuracy of 62.61%, considerably surpassing PointOBB series. Remarkably, it even outperforms the SAM-powered method P2RBox by 3.57% (62.61% vs. 59.04%).

Class-wise analysis. The FCOS detector trained with labels generated by Point2RBox-v2 achieves accuracy nearly equivalent to RBox-supervised FCOS across six high-density categories: SH (86.9% vs. 87.1%), SV (79.6% vs. 79.8%), LV (76.3% vs. 79.8%), PL (88.0% vs. 89.1%), ST (82.9% vs. 84.6%), and TC (89.1% vs. 90.4%). Interestingly, these six high-density categories account for 88% of DOTA instances. By annotating these categories with points and generating RBoxes using Point2RBox-v2 while labeling the other sparse categories with RBoxes, we can significantly reduce annotation labor without sacrificing much accuracy, highlighting the valuable role our method can play.

4.2. Results on More Datasets

The results are displayed in Table 2. On more challenging DOTA-v1.5/2.0, Point2RBox-v2 presents a similar trend, 23.47%/18.15% higher than PointOBB-v2 in the pseudo-generation track. On the ship detection dataset HRSC, the gap between Point2RBox-v2 and RBox-supervised FCOS is only 2.84% (86.15% vs. 88.99%). DIOR is relatively sparse, leading to less improvement with our methods—lower than PointSAM (44.45% vs. 46.20%) but still higher than methods that do not use SAM. Our method also provides competitive performance on fine-grained datasets FAIR1M and STAR. In addition to remote sensing scenarios, we carry out experiments on SKU110K for densely packed retail scenes. Existing point-supervised methods struggle in this



Figure 5. Qualitative analysis on failed cases and overlap cases.

Modules					DOTA		HRSC	
$\mathcal{L}_{O}$	$\mathcal{L}_{W}$	$\mathcal{L}_{ss}$	$\mathcal{L}_{E}$	*CP*	E2E	FCOS	E2E	FCOS
✓					0.00	0.00	0.00	0.00
✓	✓				41.54	52.98	17.96	19.64
✓	✓	✓			46.64	54.26	18.10	22.13
✓	✓	✓	✓		49.55	61.88	78.79	83.79
	✓	✓	✓	✓	48.58	59.56	20.35	24.76
✓		✓	✓	✓	38.94	48.44	11.64	14.93
✓	✓	✓		✓	47.08	55.05	19.58	21.78
✓	✓	✓	✓	✓	**51.00**	**62.61**	**82.67**	**86.15**

Table 8. Ablation with incremental addition of modules.

16	$K=24$	32	1.2	$\beta=1.6$	2.0
50.87	**51.00**	48.08	48.14	51.00	**51.33**

Table 9. Ablation with K and β in edge loss on DOTA (E2E).

σ	Point2RBox		PointOBB-v2		Point2RBox-v2	
	DOTA	**HRSC**	**DOTA**	**HRSC**	**DOTA**	**HRSC**
0%	40.27	79.40	44.85	-	62.61	86.15
10%	39.60	78.81	42.30	-	61.58	85.76
30%	38.42	78.28	38.46	-	60.31	85.71

Table 10. Ablation with the inaccuracy in point annotations.

case, whereas Point2RBox-v2 achieves performance on par with HBox-supervised H2RBox (65.64% vs. 57.15%).

4.3. Ablation Studies

Tables 3-10 display the ablation studies on DOTA-v1.0 and HRSC. "E2E" denotes end-to-end training; "FCOS" denotes two-stage training (i.e. generating pseudo labels to train FCOS). The final values adopted are highlighted in gray.

Weight of each loss. Tables 3-5 determine the weights of the proposed losses. Based on these experiments, the weights $(w_{O}, w_{W}, w_{E}, w_{ss})$ are set to $(10, 5, 0.3, 1)$.

Proportion of augmented views. Table 7 studies the proportion between rotation, flip, and scale. The results are reported with two-stage training (FCOS). Based on the results, the proportion is set to 68%, 7%, and 25%.

Incremental addition of modules. Table 8 demonstrates the constraints from Gaussian and Voronoi achieve an accuracy of 52.98% on DOTA. Adding consistency loss and edge loss further boosts it to 54.26% and 61.88%, respectively, whereas the improvement from copy-paste is 0.73%. We also demonstrate the impact of omitting each core loss.

Edge loss parameters. We set $K = 24$ and $\beta = 1.6$ as they are observed to discern the correct edges during code development. Table 9 provides a more precise ablation.

Annotation inaccuracy. We offset the annotated points by a noise from the uniform distribution $[-\sigma H, +\sigma H]$, where H is the height of objects. Table 10 shows that the AP_{50} of Point2RBox-v2 decreases by less than 3% when noise is added to point annotations, demonstrating the robustness of the proposed learning mechanisms.

4.4. More Discussions

The qualitative analysis on the failed/overlap cases is shown in Fig. 5. **1) Failed cases.** Although our method performs well overall, it struggles with certain categories that are sparse and not constrained by other objects. **2) Overlap cases.** Minimizing overlap as a soft constraint during training does not entirely eliminate overlap. Once trained, the model remains robust to some overlap during inference.

5. Conclusion

This paper introduces Point2RBox-v2, a point-supervised oriented object detector that effectively leverages the arrangement and layout of instances. We propose the integration of Gaussian overlay and Voronoi tessellation to constrain the size and rotation of instances based on their spatial relationships. Additionally, by incorporating self-supervised consistency loss, edge loss, and copy-paste augmentation, the accuracy of the model is further enhanced.

Experiments yield the following observations: **1)** The integration of Gaussian and Voronoi concepts effectively harnesses the spatial layout of objects, significantly enhancing point-supervised OOD. **2)** Point2RBox-v2 demonstrates exceptional performance in densely packed scenes (see Fig. 2), where existing methods struggle. **3)** Our method does not require priors (i.e. pre-trained SAM or one-shot examples) and is applicable to both end-to-end and pseudo-generation modes. **4)** It advances the state of the art by a large amount, achieving 62.61%, 86.15%, and 34.71% on the DOTA-v1.0, HRSC, and FAIR1M datasets, respectively.

Limitations. The gap between Point2RBox-v2 and the RBox-supervised OOD is still huge in terms of sparse categories (i.e. BR/SBF) since little constraint can be obtained from the layout between them when the objects are sparse.

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